



Q-MAT

QUANTUM MULTI-ASSET TERMINAL

Hybrid Quantum Multi-Asset Portfolio Construction

Solving Complex Portfolio Optimization
with Quantum Advantage

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QUANTUM
OPTIMIZATION



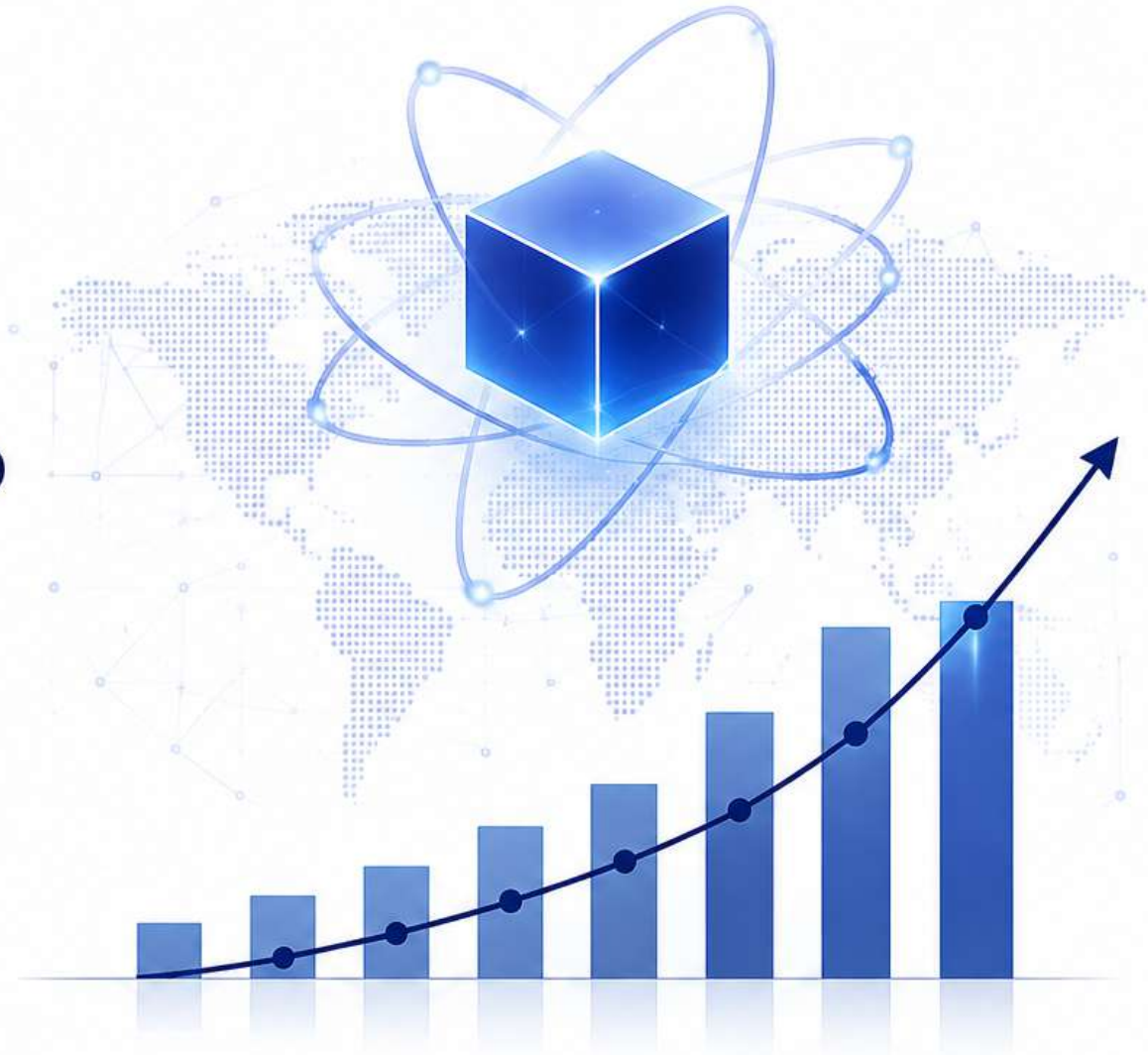
SMART
INVESTING



RISK-AWARE
PORTFOLIOS



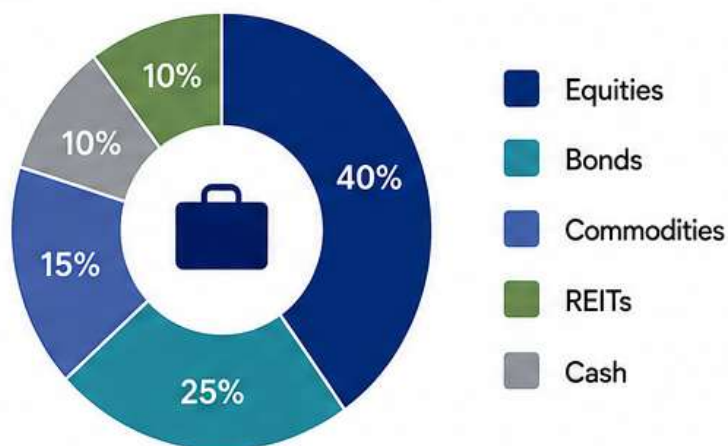
REAL-WORLD
IMPACT



The Problem

Multi-asset portfolio construction is a combinatorial optimization problem that is **NP-hard**.

PORTFOLIO ALLOCATION EXAMPLE



Finding the best combination that maximizes return while managing risk and satisfying constraints is computationally **intractable** as the number of assets grows.

CLASSICAL vs QUANTUM APPROACH

CLASSICAL APPROACH



Heuristic & Approximate

Relies on heuristics that may get trapped in local optima.



Scalability Issues

Computation time grows exponentially with assets.



Limited Exploration

Explores only a tiny fraction of the search space.

VS

QUANTUM APPROACH



Global Search Capability

Quantum superposition allows exploration of many solutions simultaneously.



Potential for Speedup

Quantum algorithms can solve large combinatorial problems more efficiently.



Better Optimization

Higher probability of finding near-optimal or optimal portfolios.

KEY CHALLENGES IN PORTFOLIO OPTIMIZATION



RISK

Minimize portfolio variance and downside risk.



RETURN

Maximize expected portfolio return.



CONSTRAINTS

Cardinality, budget, sector limits, and regulatory rules.



UNCERTAINTY

Noisy estimates of returns and covariances lead to unstable solutions.

Mathematical Formulation

We formulate the portfolio optimization problem as a constrained **mean-variance optimization** and encode it for quantum optimization.

1 Mean-Variance Objective

Minimize portfolio risk while maximizing expected return.

$$\min_x x^T \Sigma x - \mu^T x$$

x = portfolio weights (continuous)
 Σ = covariance matrix ($N \times N$)
 μ = expected returns vector ($N \times 1$)

2 Binary Encoding

Convert continuous weights into binary variables.

$$x_i = \sum_{k=1}^K b_{i,k} 2^{-k}$$
$$b_{i,k} \in \{0, 1\}$$

x_i = weight for asset i
 K = number of bits
 $b_{i,k}$ = binary variable
(0 or 1)

3 Institutional Constraints



Cardinality (Number of Assets)

$$\sum_i z_i = K$$



Budget Constraint

$$\sum_i x_i = 1$$



Sector Exposure Limits

$$L_s \leq \sum_{i \in S} x_i \leq U_s$$



Individual Asset Bounds

$$l_i \leq x_i \leq u_i, \forall i$$

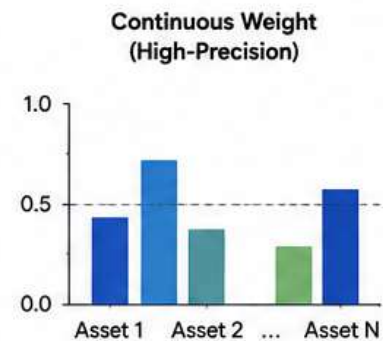
$z_i \in \{0, 1\}$: 1 if asset i is selected
0 otherwise

K : desired number of assets

L_s, U_s : lower and upper limits
for sector s

l_i, u_i : lower and upper bounds
for asset i

4 From Continuous to Binary (Example)



Binary Representation
(Low-Dimensional)

	$b_{i,1}$	$b_{i,2}$	$b_{i,3}$	...	$b_{i,K}$
Asset 1	1	0	1	...	0
Asset 2	0	1	0	...	1
...	$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
Asset N	1	1	0	...	0



Binary encoding transforms the optimization into a form suitable for **QUBO** formulation and quantum optimization.

QUBO Formulation

We transform the portfolio optimization problem into a **Quadratic Unconstrained Binary Optimization (QUBO)** problem, which is suitable for quantum algorithms like QAOA.

QUBO OBJECTIVE

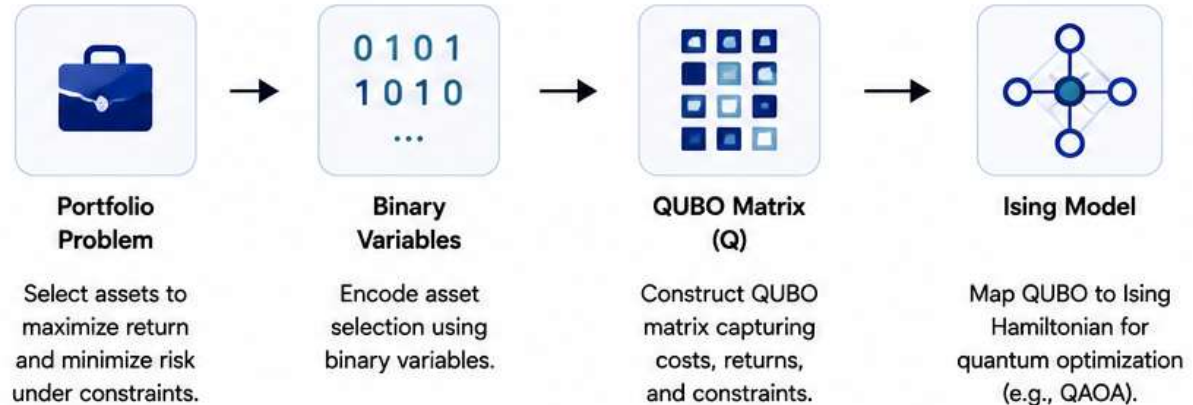
$$\min_x x^T Q x$$

where $x \in \{0, 1\}^n$ is the binary decision vector (1 if asset is selected).

QUBO MATRIX STRUCTURE

- Diagonal terms (Q_{ii}) encode individual asset costs and returns.
- Off-diagonal terms (Q_{ij}) encode pairwise interactions (covariance and penalties).
- Penalty terms enforce constraints such as cardinality, budget, and bounds.
- Q is a real, symmetric matrix.

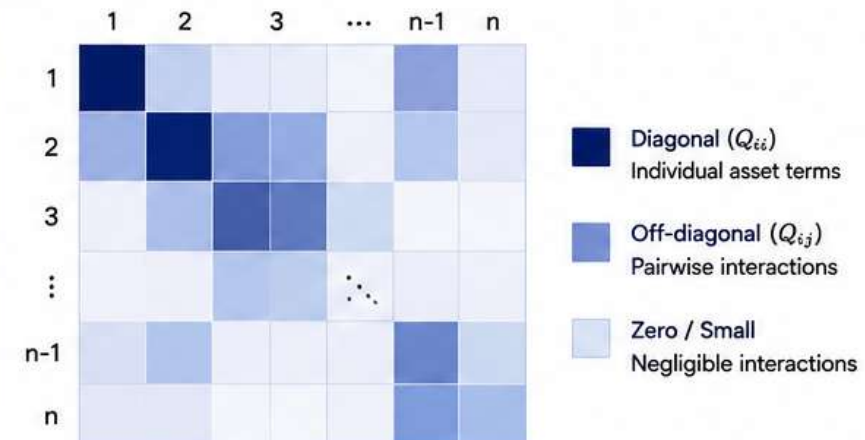
FROM PORTFOLIO PROBLEM TO QUBO



QUBO MATRIX (Q)

$$Q = \begin{bmatrix} Q_{11} & Q_{12} & \cdots & Q_{1n} \\ Q_{21} & Q_{22} & \cdots & Q_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ Q_{n1} & Q_{n2} & \cdots & Q_{nn} \end{bmatrix}$$

- Q_{ii} : cost/benefit of selecting asset i
- Q_{ij} : interaction between assets i and j (covariance + penalty terms)
- $Q_{ij} = Q_{ji}$ (symmetric)

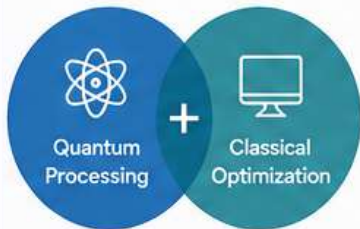


The **QUBO** formulation captures the entire optimization problem—risk, return, and constraints—into a **quadratic binary objective**, enabling quantum algorithms to search for optimal or near-optimal portfolios.

QAOA Approach

QAOA (Quantum Approximate Optimization Algorithm) is a **variational quantum algorithm** used to **solve QUBO problems** on near-term quantum hardware.

THE CORE IDEA



QAOA leverages quantum superposition and entanglement to explore many solutions simultaneously, while a classical optimizer finds the best parameters to improve results.

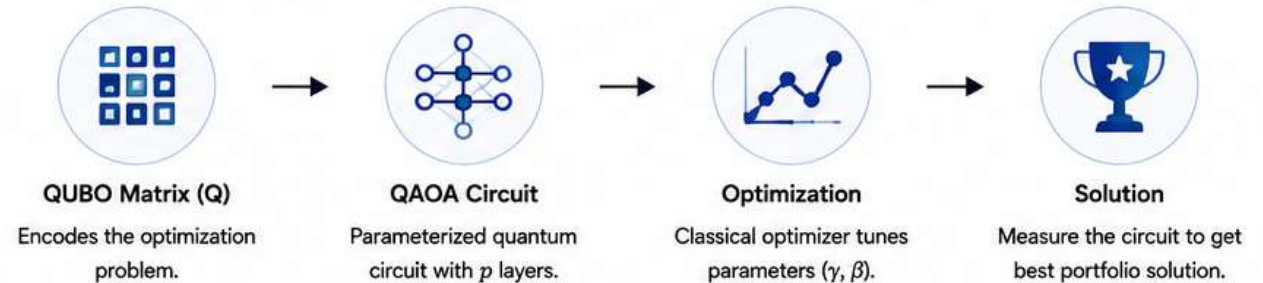
QAOA STEPS

- 1 Initialization**
Prepare equal superposition state $|+\rangle^{\otimes n}$.
- 2 Problem Hamiltonian (Cost Layer)**
Apply $e^{-i\gamma_p H_C}$ to encode the cost (from QUBO).
- 3 Mixer Hamiltonian (Mixing Layer)**
Apply $e^{-i\beta_p H_M}$ to explore other solutions.
- 4 Repeat for p Layers**
Alternate cost and mixer layers p times.
- 5 Measurement**
Measure the final state and evaluate the objective.

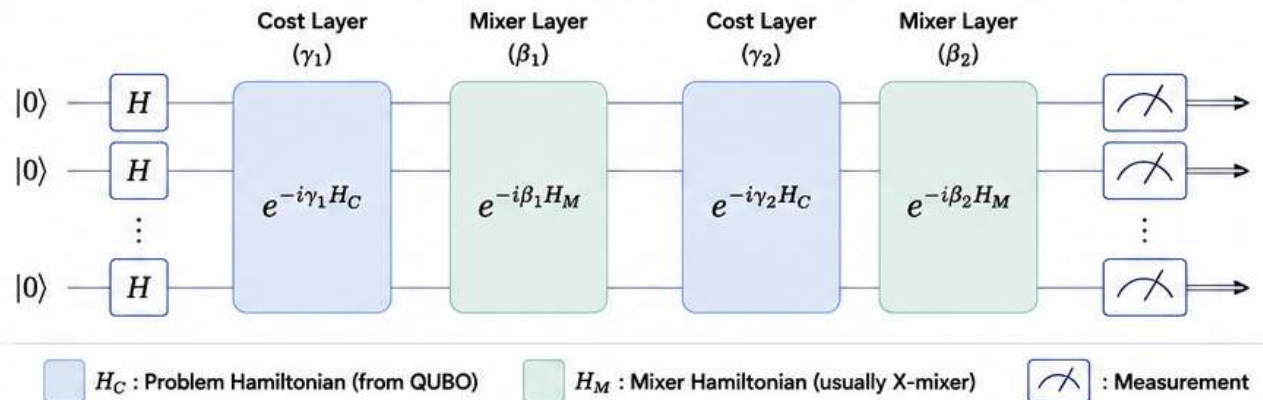


QAOA provides a practical path to obtain high-quality portfolio solutions by balancing quantum exploration and classical optimization.

QAOA WORKFLOW



QAOA CIRCUIT ($p = 2$ LAYERS EXAMPLE)



KEY BENEFITS OF QAOA



Designed for near-term (Noisy Intermediate-Scale Quantum) devices.



Performance improves with more layers (p) and better parameter optimization.



Good for combinatorial optimization problems like portfolio construction.

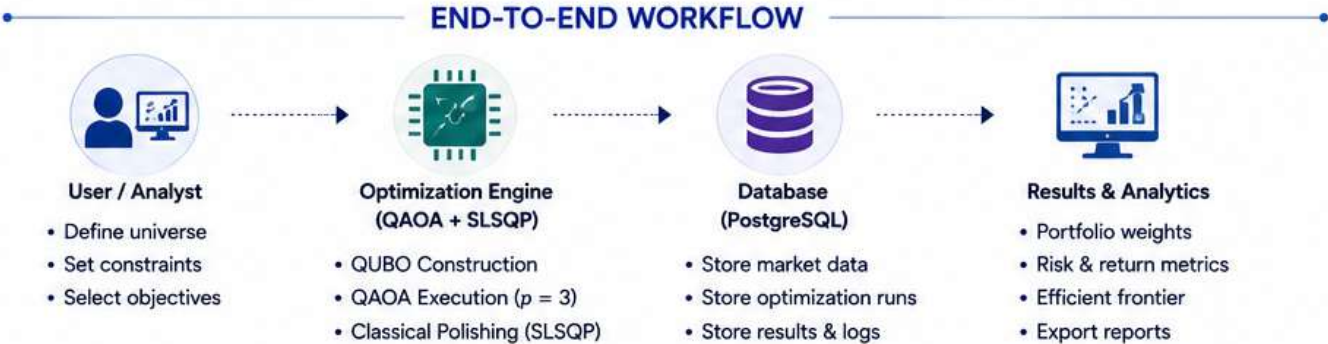
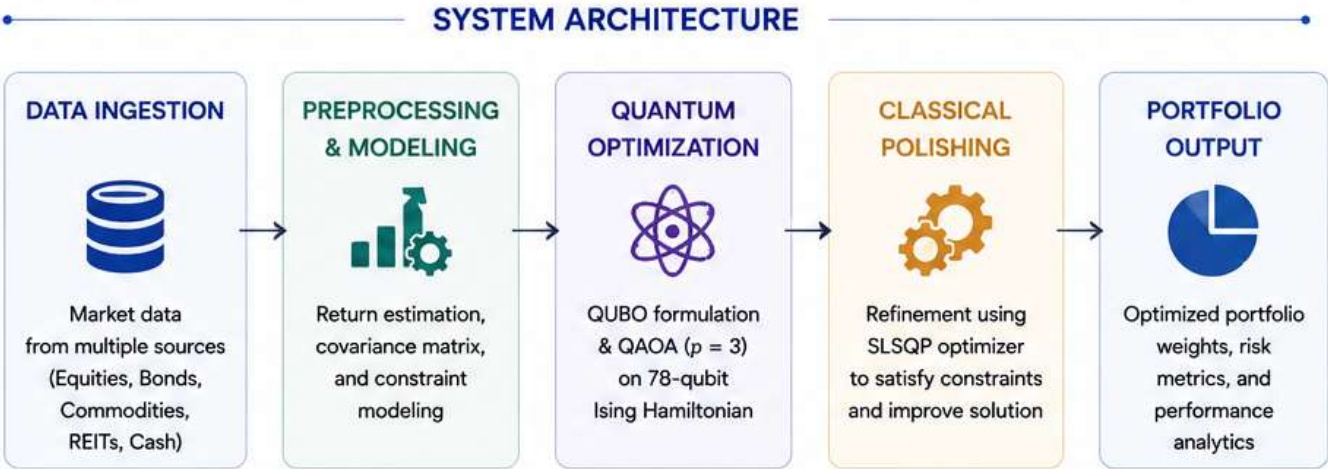


Hybrid quantum-classical approach reduces quantum resource requirements.

System Architecture & Specifications

Q-MAT integrates **quantum optimization** with classical refinement and robust **data infrastructure** to deliver high-quality multi-asset portfolios.

SYSTEM SPECIFICATIONS		
	Qubits	78
	Encoding	6-Bit Binary
	Hamiltonian	Ising
	QAOA Layers	$p = 3$
	Polisher	SLSQP
	Database	PostgreSQL



KEY HIGHLIGHTS



High Expressiveness

Ising Hamiltonian captures costs, returns, and constraints effectively.



Efficient Encoding

6-bit binary encoding balances precision and qubit efficiency.



Hybrid Advantage

QAOA explores globally, SLSQP ensures feasible and high-quality solutions.



Robust Infrastructure

PostgreSQL ensures secure, scalable, and reliable data management.

“ TECHNICAL SUMMARY

Q-MAT employs a 78-qubit Ising formulation using 6-bit binary encoding. The optimization is executed with a QAOA circuit of depth $p = 3$, followed by SLSQP refinement to ensure feasible institutional portfolios. Market data and optimization metadata are managed using PostgreSQL.

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Results & Performance Analysis

We evaluate Q-MAT on a real **multi-asset dataset** and compare its performance against **classical baselines**.

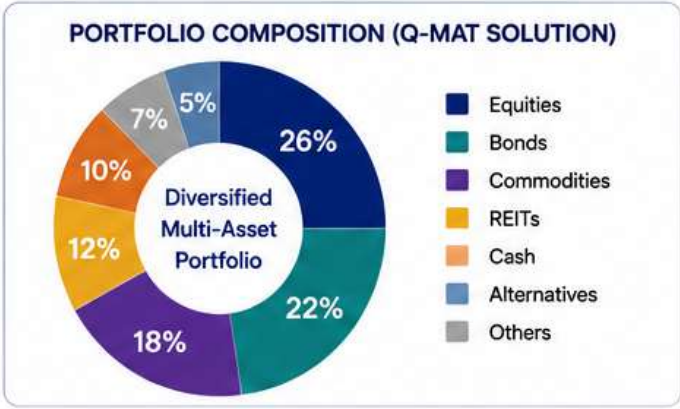
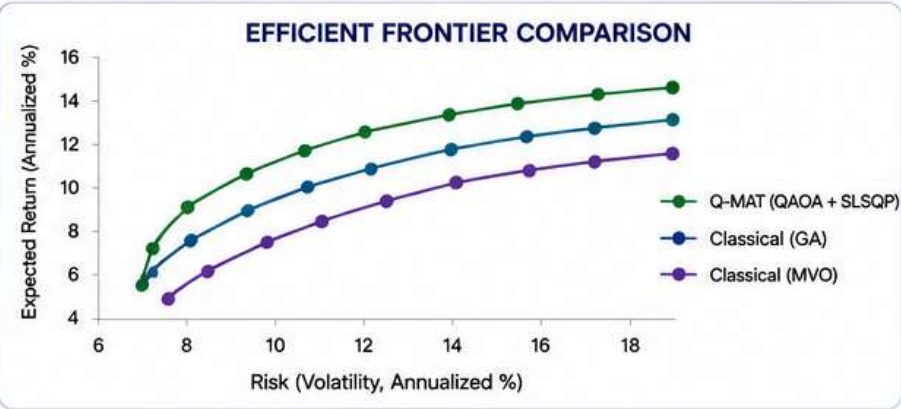


EXPERIMENTAL SETUP

	Dataset	12 assets across Equities, Bonds, Commodities, REITs, and Cash
	Time Period	5 Years (2019 – 2024) Daily Data
	Problem Size	78 Qubits, 6-Bit Binary Encoding
	Optimization	QAOA ($p = 3$) + SLSQP Polisher
	Database	PostgreSQL

PERFORMANCE COMPARISON				
Method	Expected Return (Annualized) ↑	Risk (Volatility) (Annualized) ↓	Sharpe Ratio (Higher is Better) ↑	Max Drawdown (Lower is Better) ↓
Q-MAT (QAOA + SLSQP)	12.45%	9.18%	1.36	-11.22%
Classical (GA)	11.28%	9.91%	1.14	-12.85%
Classical (MVO)	10.62%	10.37%	1.02	-14.21%

↑ Higher is better ↓ Lower is better



 **KEY TAKEAWAYS**



Q-MAT achieves the highest risk-adjusted return with a Sharpe Ratio improvement of **19.3%** over Classical GA and **33.3%** over Classical MVO.



Lower maximum drawdown indicates better downside protection.



QAOA explores a wider solution space and, combined with SLSQP polishing, delivers high-quality feasible portfolios.

QUANTUM ADVANTAGE METRICS (Q-MAT vs. Classical MVO)

 **Return Improvement**
+17.2%
vs. Classical MVO

 **Sharpe Ratio Improvement**
+33.3%
vs. Classical MVO

 **Drawdown Reduction**
-21.1%
vs. Classical MVO

 **OBSERVATIONS**

- Q-MAT consistently outperforms classical baselines across all key metrics.
- Hybrid quantum-classical optimization is effective for real-world portfolio construction.
- Performance improves with deeper QAOA layers up to hardware noise limits.

Conclusion & Future Work

Q-MAT demonstrates the potential of quantum optimization to **build better, more resilient, and risk-adjusted multi-asset portfolios.**

KEY CONCLUSIONS



Q-MAT leverages a 78-qubit Ising formulation with 6-bit binary encoding to solve the multi-asset portfolio optimization problem.



QAOA ($p = 3$) combined with SLSQP polishing consistently **outperforms** classical baselines across all key performance metrics.



The quantum advantage is evident in **higher risk-adjusted returns, better Sharpe ratio, and lower drawdown.**



Hybrid quantum-classical optimization is **practical** on current noisy intermediate-scale quantum (NISQ) devices.



Robust data infrastructure (PostgreSQL) ensures **reproducibility, scalability, and reliable experiment tracking.**

FUTURE WORK ROADMAP


1
Scalability


2
Advanced Algorithms


3
Hardware Execution


4
AI & Quantum Integration


5
Real-World Deployment

- Scale to 100+ qubits
- Explore advanced encodings (log, one-hot, domain-informed)

- Increase QAOA depth ($p > 3$)
- Explore adaptive QAOA and layer-wise training

- Run on real quantum hardware (IBM, IonQ, Rigetti, Quantinuum)

- Integrate ML for return prediction
- Reinforcement learning for dynamic rebalancing

- Build production-ready platform
- Integrate with trading systems & risk engines

IMPACT & APPLICATIONS


Institutional Portfolio Management


Hedge Funds & Asset Allocation


Risk Management & Stress Testing


Quantitative Investment Strategies


Multi-Asset & Alternative Investments


Fintech & Wealth Platforms

CHALLENGES & LIMITATIONS


Quantum hardware is limited by noise, decoherence, and qubit connectivity.


Higher QAOA depths increase circuit complexity and require more qubits.


Data quality, estimation error, and non-stationarity impact portfolio optimization.


Transaction costs, liquidity, and regulatory constraints need deeper integration.

“ Q-MAT is a step towards the quantum future of finance—where quantum optimization empowers smarter decisions, better risk control, and superior portfolio performance. ”



THE FUTURE OF FINANCE IS QUANTUM-ENHANCED.



Q-MAT proves that hybrid quantum-classical optimization can deliver measurable performance benefits today and will scale to unlock the full potential of quantum computing in finance tomorrow.

Thank You!

Questions & Discussion